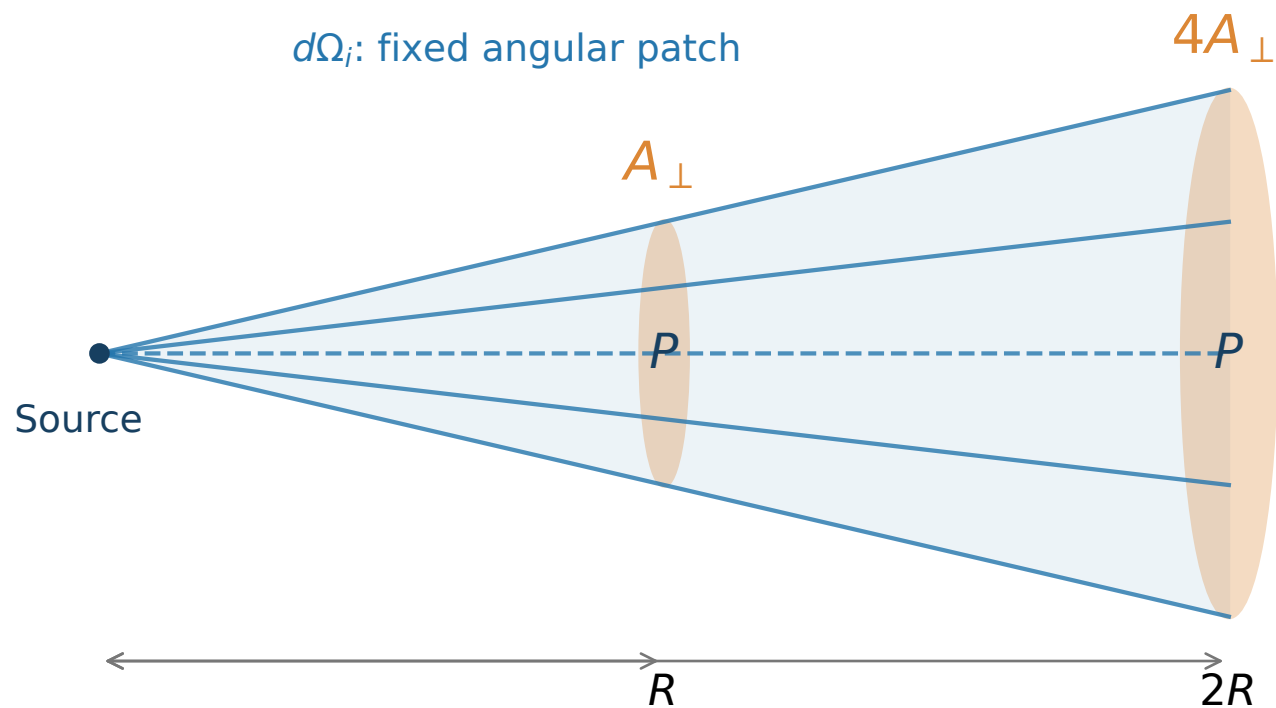
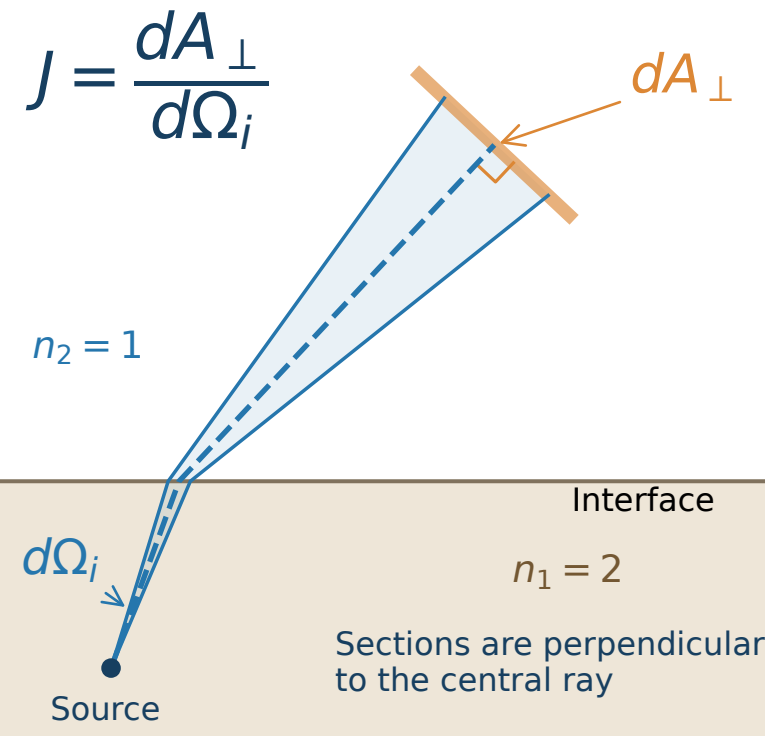


A ray tube: a narrow bundle of neighboring rays



Same lossless medium: $I = P/A_{\perp}$, $|E| \propto A_{\perp}^{-1/2}$
 Distance x2 $\rightarrow$ Area x4 $\rightarrow$ Intensity /4 $\rightarrow$ Field /2

Refraction changes how the bundle spreads



Geometric spreading; Fresnel transmission and absorption are separate
 Illustration: 2D views of 3D bundles; angular width enlarged